EXAMINATION NO.:_____

THE MALAWI NATIONAL EXAMINATIONS BOARD

2012 MALAWI SCHOOL CERTIFICATE OF EDUCATION

BIOLOGY

Subject Number: M022/I

Tuesday, 3 July

Time Allowed: 2 h 30 mins
8:30 – 11:00 am

PAPER I
(100 marks)

Theory

1. **This paper contains 12 pages. Please check.**
2. Before you begin, **fill** in your **Examination Number** at the top of the question paper and on all other sheets.
3. This paper contains sections **A**, **B** and **C**. Answer **all** questions in all the sections. Some can be answered quickly, but others require considerable thought and may take longer.
4. Write your answers on the question paper in the spaces provided. The maximum number of marks for each answer is indicated against each question.
5. In the table provided on this page, **tick** against the question number you have answered.
6. You should hand in your question paper to the invigilator when time is called to stop writing.

Question Number	Tick if answered	Do not write in these columns	
1			
2			
3			
4			
5			
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8			
9			
10			
11			
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15			
16			

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Section A (20 marks)

Answer **all** questions in this section.

1. a. State any **two** ways in which vectors transmit diseases.

_____

_____

(2 marks)

b. Mention any **two** signs of athletes foot.

_____

_____

(2 marks)

2. **Figure 1** shows part of a leaf. Use it to answer the questions that follow.

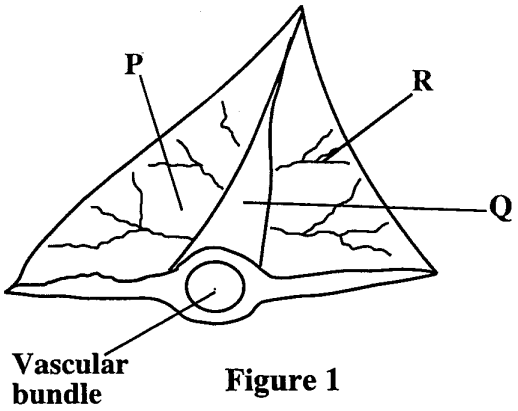

Figure 1

a. Name the part marked **Q.**

_____

(1 mark)

b. Explain the functions of **P** and **R** in relation to photosynthesis.

_____

_____

_____

_____

_____

(4 marks)

Continued/...

3. **Figure 2** shows a stage during meiosis. Use it to answer questions that follow.

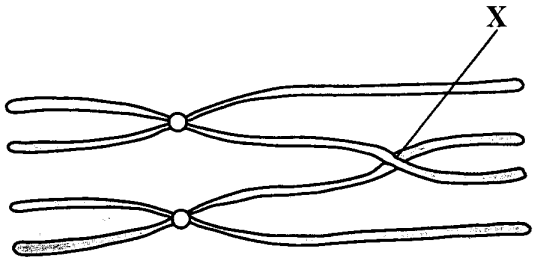

Figure 2

a. Name the process taking place at part **X**.

_____

_____

(1 mark)

b. Name any **one** organ in the human body in which process **X** takes place.

_____

_____

(1 mark)

c. Explain the importance of the process taking place at **X**.

_____

_____

_____

(2 marks)

4. **Figure 3** shows a plant tissue obtained from a stem. Use it to answer the questions that follow.

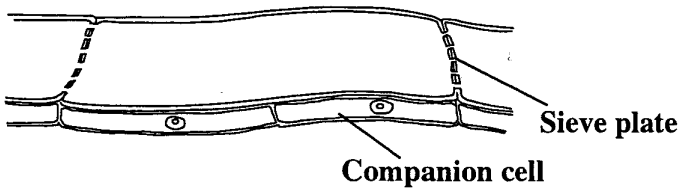

Figure 3

a. (i) Identify the tissue.

_____

(1 mark)

Continued/...

4. **(Continued)**

(ii) Give **two** reasons to support your answer in **a(i)**.

_____

_____

(2 marks)

b. Explain **two** adaptations of the tissue to its function.

_____

_____

_____

(4 marks)

Section B (60 marks)

Answer **all** the questions in this section.

5. A man with blood group **A** was married to a woman with blood group **B**. Their first born son was of blood group **O** while their second born son was of blood group **AB**.

a. Give the genotypes of the two parents.

_____

_____

(2 marks)

b. Explain how the genotype of the second born son came about.

_____

_____

_____

(3 marks)

6. When a grey cock was mate with a grey hen, grey and white chicks were produced.

a. Using **G** for grey colour and **g** for white colour draw a generic diagram to determine the genotype of he offspring.

(3 marks)

Continued/....

6. (Continued)

b. Give the genotypic ratio of the chicks.

_____

_____

(2 marks)

c. If the parents produced 12 chicks, how many were white? Show your working.

(2 marks)

7. **Figure 4** shows one of the examples of natural selection in action. Use it to answer questions that follow.

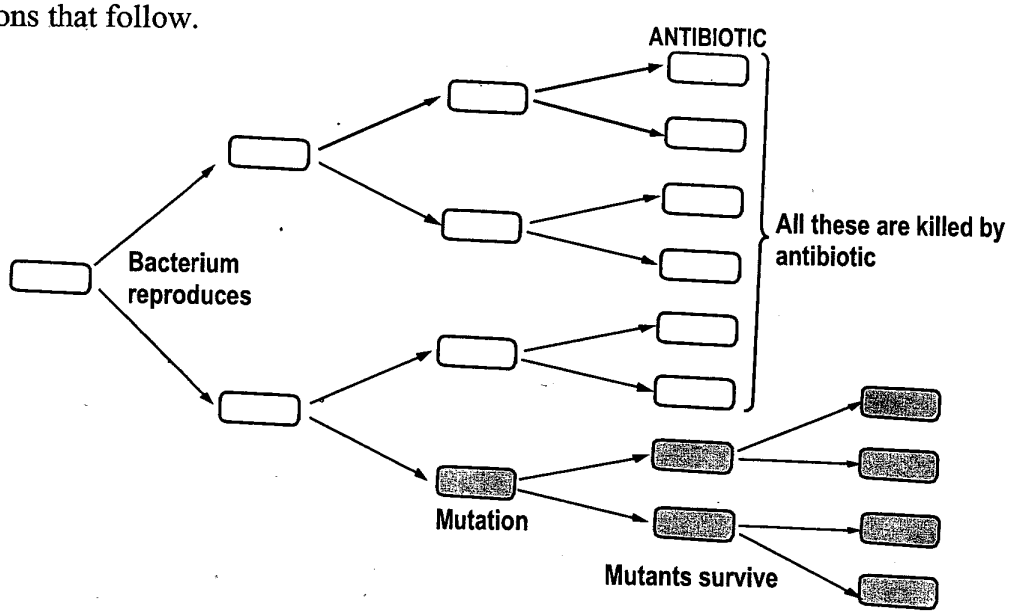

Figure 4

a. (i) Name the example of natural selection shown.

_____

(1 mark)

Continued/..

7. a. **(Continued)**

(ii) Describe what led to the survival of some of the bacteria while the others got killed.

_____

_____

_____

(2 marks)

b. What will happen to the antibiotic after sometime?

_____

_____

(1 mark)

8. **Figure 5** shows one of the nutrient cycles in nature. Use it to answer the questions that follow.

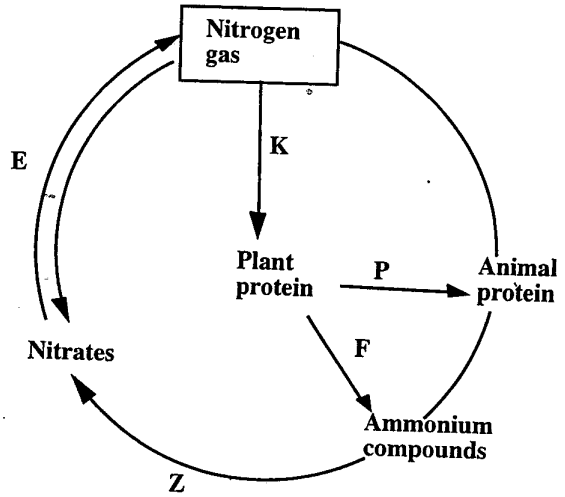

Figure 5

a. Name the processes taking place at **F**, **Z** and **P**.

F: _____

P: _____

Z: _____

(3 marks)

b. Explain how the process at **K** takes place.

_____

_____

_____

(3 marks)

Continued/...

8. (Continued)

c. Under what conditions does the process at **E** take place?

_____

_____

(1 mark)

9. **Figure 6** is a diagram of male reproductive system. Use it to answer questions that follow.

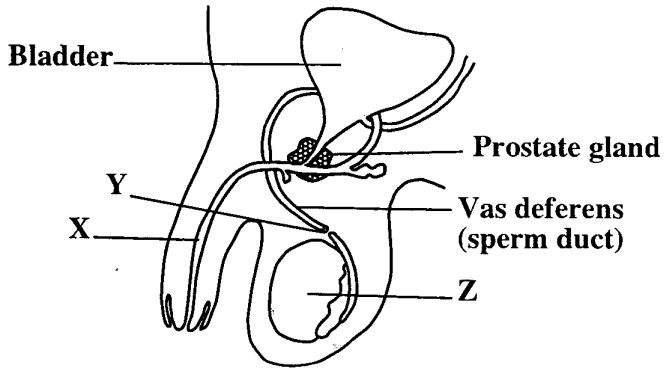

Figure 6

a. Name the parts marked **X** and **Z**.

X _____ **Z** _____

(2 marks)

b. Mention the contraceptive method shown at **Y**.

_____

(1 mark)

c. Mention any **two** advantages of using this contraceptive method.

_____

_____

(2 marks)

10. a. Mention any **two** effects of smoking on human health.

_____

_____

(2 marks)

b. State any **two** effects of exercise on breathing.

_____

_____

(2 marks)

Continued/...

10. (Continued)

c. Mention any **two** adaptations of alveoli to their function.

_____

_____

_____

(2 marks)

11. a. (i) Name the **main** nitrogenous waste excreted by kidneys.

_____

(1 mark)

(ii) Describe how the nitrogenous waste mentioned in **11a(i)** is formed.

_____

_____

_____

(3 marks)

b. Name **two** substances found in blood plasma that are **not** found in the urine of a healthy person.

_____

_____

(2 marks)

12. **Figure 7** is a diagram of the human heart. Use it to answer questions that follow.

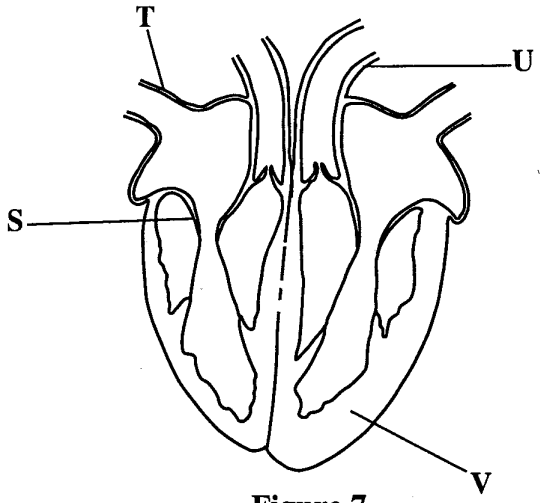

Figure 7

a. Name the part marked **S**.

_____

(1 mark)

Continued/...

12. (Continued)

b. Give **two** differences between the blood flowing through vessels **T** and **U**.

(i) _____

(ii) _____

(2 marks)

c. Explain how part **V** helps to create systolic pressure.

_____

_____

_____

(3 marks)

13. a. State any **two** adaptations of a villus for food absorption.

_____

_____

(2 marks)

b. If the liver failed to produce bile, what two effects could this have on digestion?

_____

_____

_____

(2 marks)

c. Explain why the level of insulin increases in the blood soon after a meal of carbohydrates.

_____

_____

_____

(2 marks)

Continued/...

14. **Figure 8** shows capillary bed. Use it to answer questions that follow.

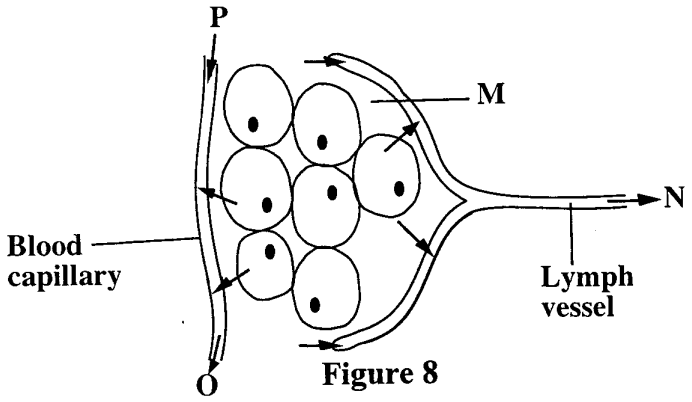

Figure 8

a. (i) Name the fluid found in **M**.

_____

(1 mark)

(ii) Name any **two** substances found in **M** that are used by the body.

_____

_____

(2 marks)

b. Name **two** adaptations for each of the following to their functions:

(i) blood capillaries

_____

_____

(2 marks)

(ii) lymph vessels

_____

_____

(2 marks)

c. Which letter represents the arterial end of the capillary bed?

_____

(1 mark)

Continued/...

Section C (20 marks)

Essay Questions

Answer **all** questions in this section.

15. Explain any **five** ways in which breast feeding is important.

(10 marks)

Continued/...

16. Explain how one can estimate the population of fish in a pond. Your answer should be in an essay form.

(10 marks)

END OF QUESTION PAPER

NB: This paper contains 12 pages.